Coherence of light

Coherence

- ➔ Coherence is a notion of regularity
- ➔ Coherence means nonzero correlation, and well-defined phase relationship between two spatio-temporal points
- ➔ Coherence means nonzero correlation between two temporal points at a given spatial point or between two spatial points at a given time
- ➔ For light, nonzero correlation means definite (not random) phase relation
- ➔ Test of coherence is through interference experiment - stable interference pattern means existence of coherence between interfering waves
- ➔ Quantitative measure of coherence is through visibility (contrast) of the interference fringe pattern

Drill of an unit of soldiers - an analogy to coherence

https://www.youtube.com/watch?v=kZF7QC3tsJY

Coherence

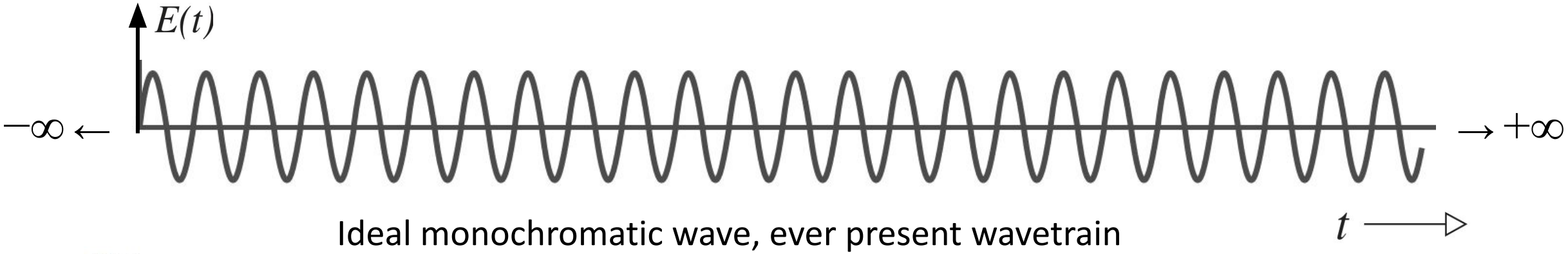
- Coherence - nonzero phase correlation, well-defined phase relationship
  - Temporal (longitudinal) coherence - correlation between two temporal or two spatial points along the direction of propagation - related to spectral width of light
  - Spatial (transverse) coherence - correlation between light coming from two spatial points located perpendicular to the direction of propagation - related to spatial extent (size) of the source

We shall first look at temporal coherence.

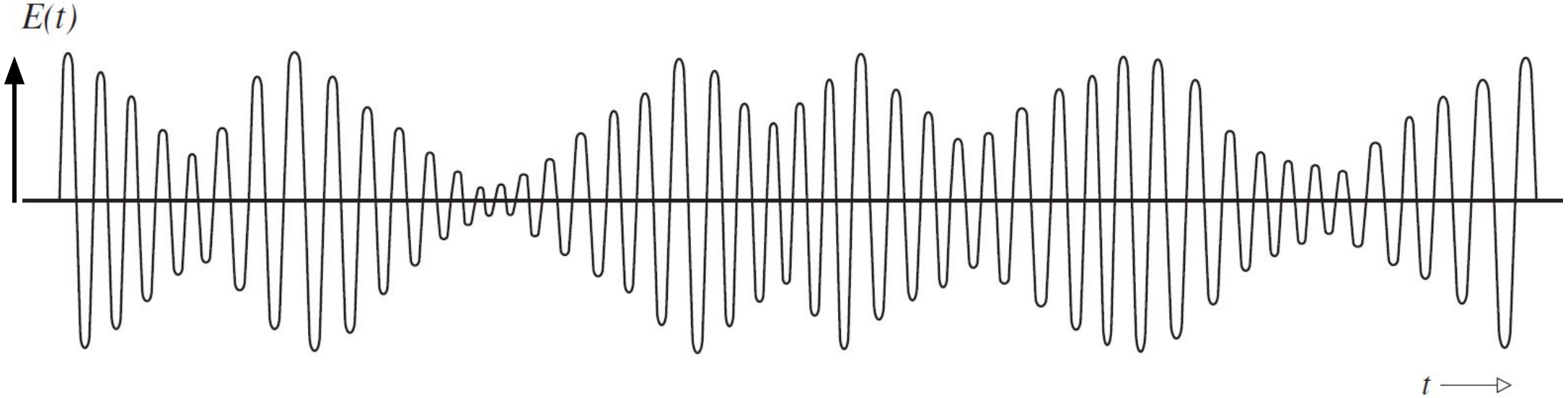
A monochromatic plane wave has complete coherence - infinite coherence time

- Consider electric field of a plane wave: $E(x,t) = E_0 \cos(\omega t - kx + \phi)$
- At a fixed position, say at $x = 0$, $\longrightarrow E(t) = E_0 \cos(\omega t + \phi)$
- It represents a sinusoid, extended over $t = -\infty$ to $t = +\infty$
- It is an infinite wavetrain - it has no beginning and no ending - it is present forever. This of course is a mathematical idealization
- Ideally, it represents a fully coherent wave - if we know the state of the wave at any given time, we know the state of the wave at all times
- It also represents a fully monochromatic wave - only one frequency (and therefore only one wavelength) is present
- This of course does not happen in reality in nature - there is no perfectly monochromatic wave - no infinitely extended wavetrain exists in reality

An ideal monochromatic wave vs a quasi-monochromatic wave



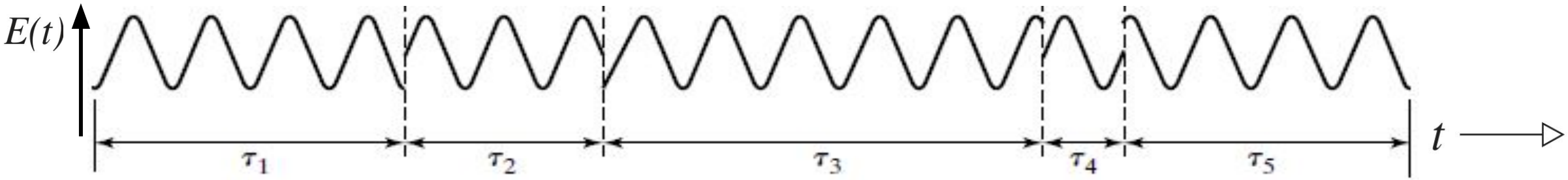
Ideal monochromatic wave, ever present wavetrain



Quasi-monochromatic wave, consists of several finite wavepackets; phase and amplitude varies slowly w.r.t optical frequency

Natural light sources emit light as bursts of finite wavetrains

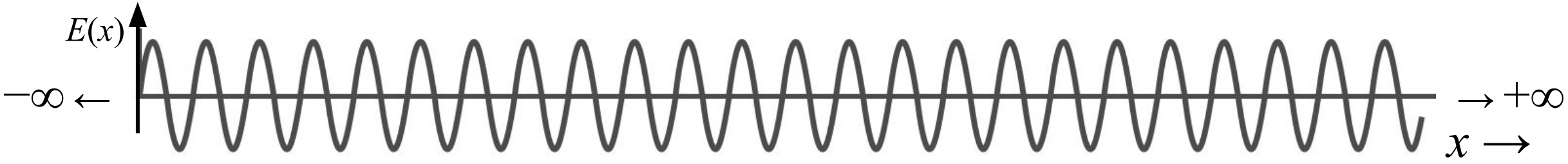
- Light emission from natural light sources are based on random processes
- Atoms are in random thermal motion, finite lifetime of emission process etc. contribute to the finite and random duration of wavetrains



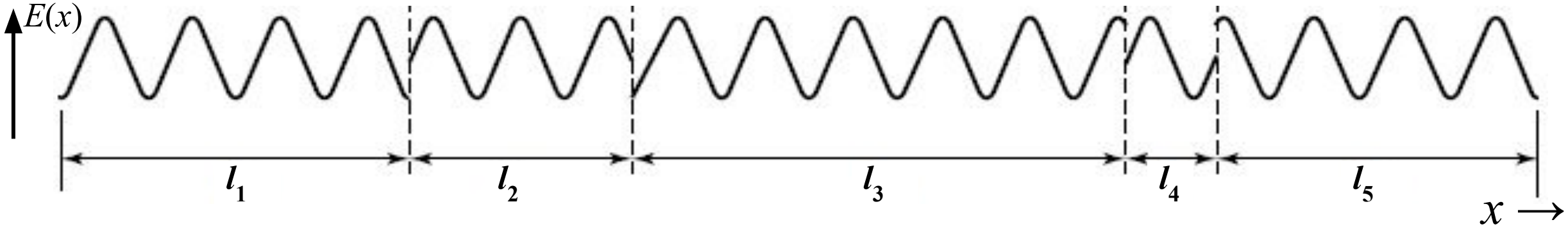
- Average duration of the wavetrain, $\tau_c = \langle \tau_i \rangle$ defines the duration over which a phase correlation exists → if we know the phase of a wave at given position at time t, we can reasonably predict the phase of the wave at the same position at a later time ($t + \Delta t$) if $\Delta t \lesssim \tau_c$
- τ_c defines the coherence time of the light source, and the emitted light is said to remain coherent for times ~τ_c

Coherence length associated with temporal (longitudinal) coherence

- We can consider similar scenario in spatial domain:
  - Electric field of wave at fixed time (say at $t = 0$): $E(x) = E_0 \cos(kx - \phi)$
  - Represents an infinitely extended wavetrain



- In reality, finite wavetrains, random duration, random length



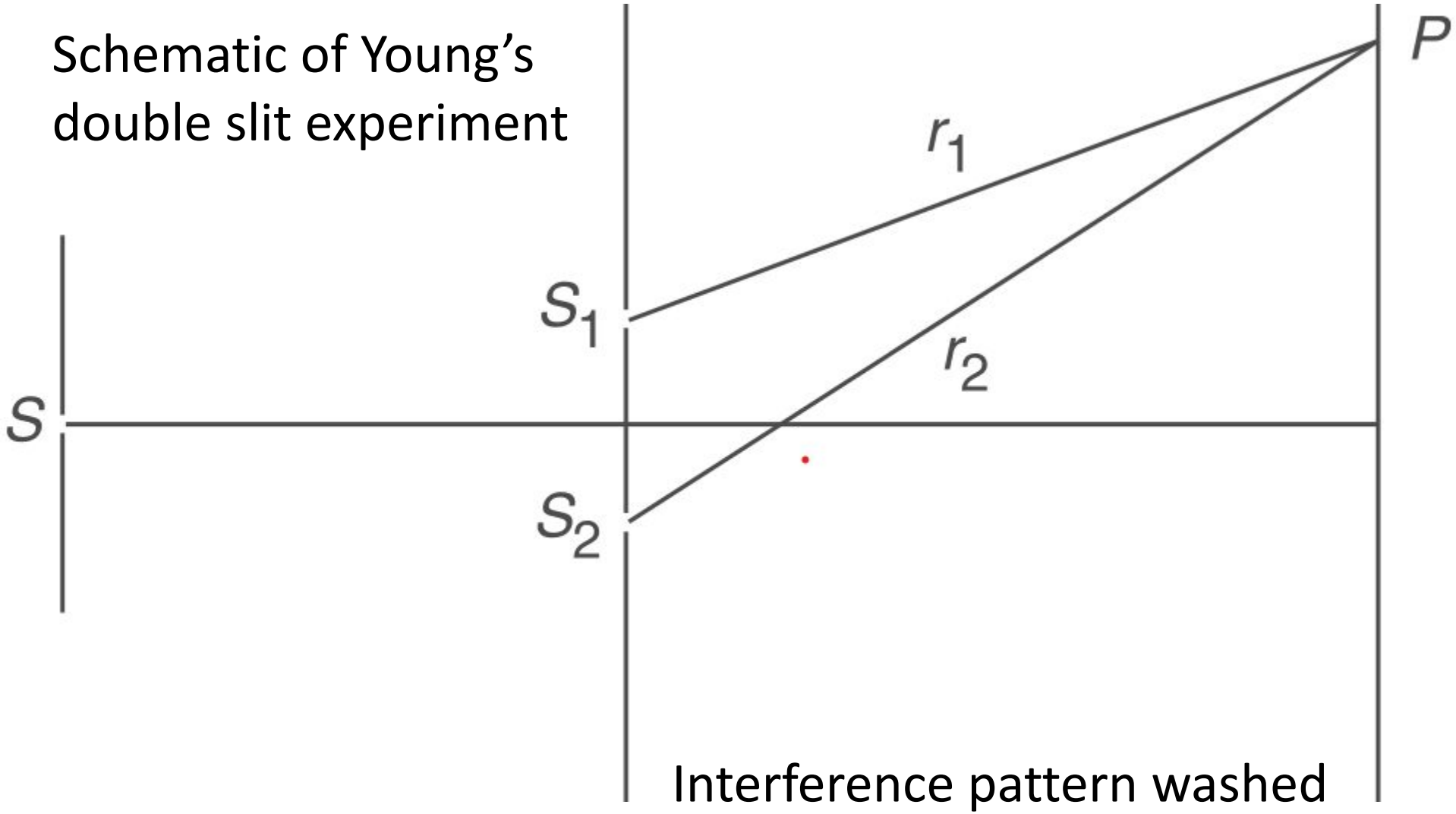
- Coherence length: $l_c = \langle l_i \rangle \equiv c\tau_c$

Coherence length associated with temporal (longitudinal) coherence

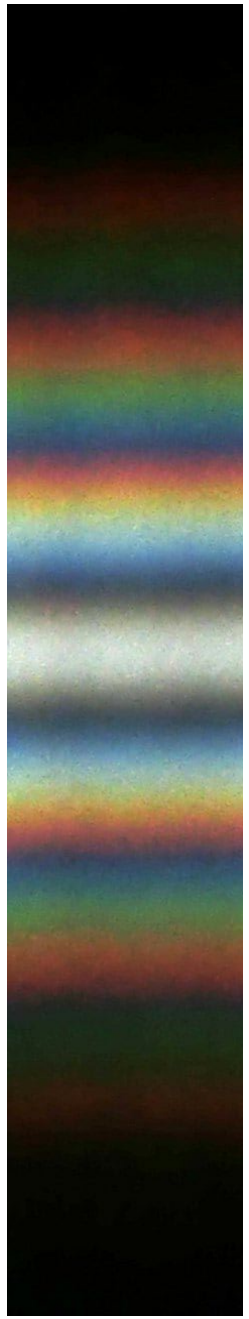
- Average length of the wavetrain, $l_c = \langle l_i \rangle \equiv c\tau_c$ defines the length over which a phase correlation exists $\longrightarrow$ if we know the phase of a wave at given time at position x, we can reasonably predict the phase of the wave at the same time at a distant position ($x + \Delta x$) if $\Delta x \lesssim l_c$
- l_c defines the coherence length of the light source, and the emitted light is said to remain coherent for a lengthscale ~l_c

Interference as a measure of coherence

Schematic of Young's double slit experiment

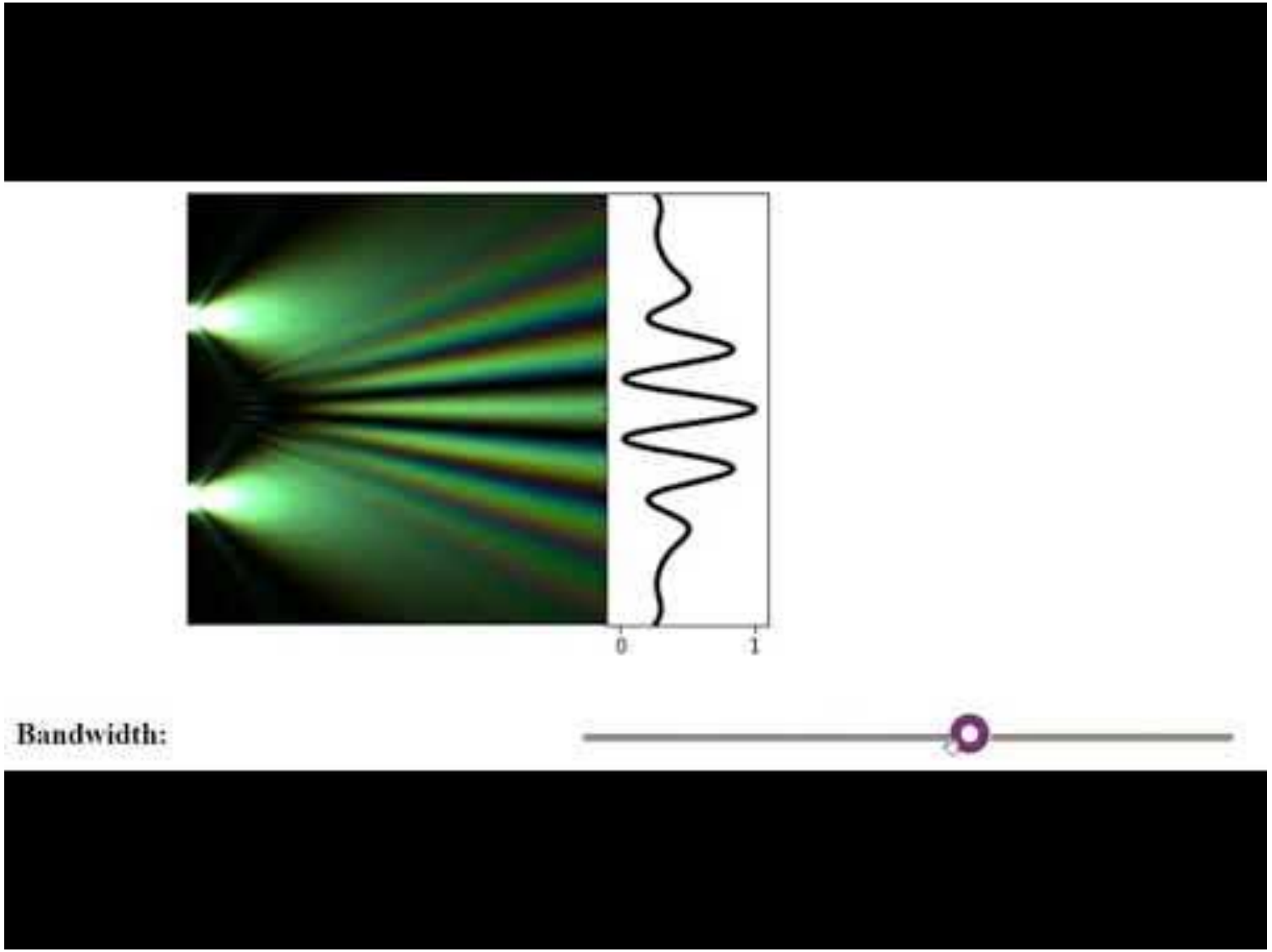


Interference pattern washed out when $|r_2 - r_1| >\sim l_c \equiv c\tau_c$



Interference as a measure of coherence

As the spectral width is increased, the transverse extent of interference pattern is reduced - lesser no. of clear fringes are seen - fringes in the peripheral region gets washed out



Effect of spectral width on intensity pattern in Young’s double slit experiment

Finite time duration of emitted wavetrains lead to spectral broadening

- The spectral width is inversely proportional to the average duration of the wavetrains $\longrightarrow$ spectral width and coherence time has a inverse relationship
- This can be establish easily by knowing that time and frequency are two conjugate Fourier variables
- Distribution of electric field amplitude in time domain, $E(t)$ is related through Fourier transform with the distribution of electric field amplitude in frequency domain, $\tilde{E}(\omega)$, i.e. $E(t)$ and $\tilde{E}(\omega)$ are Fourier transform of each other

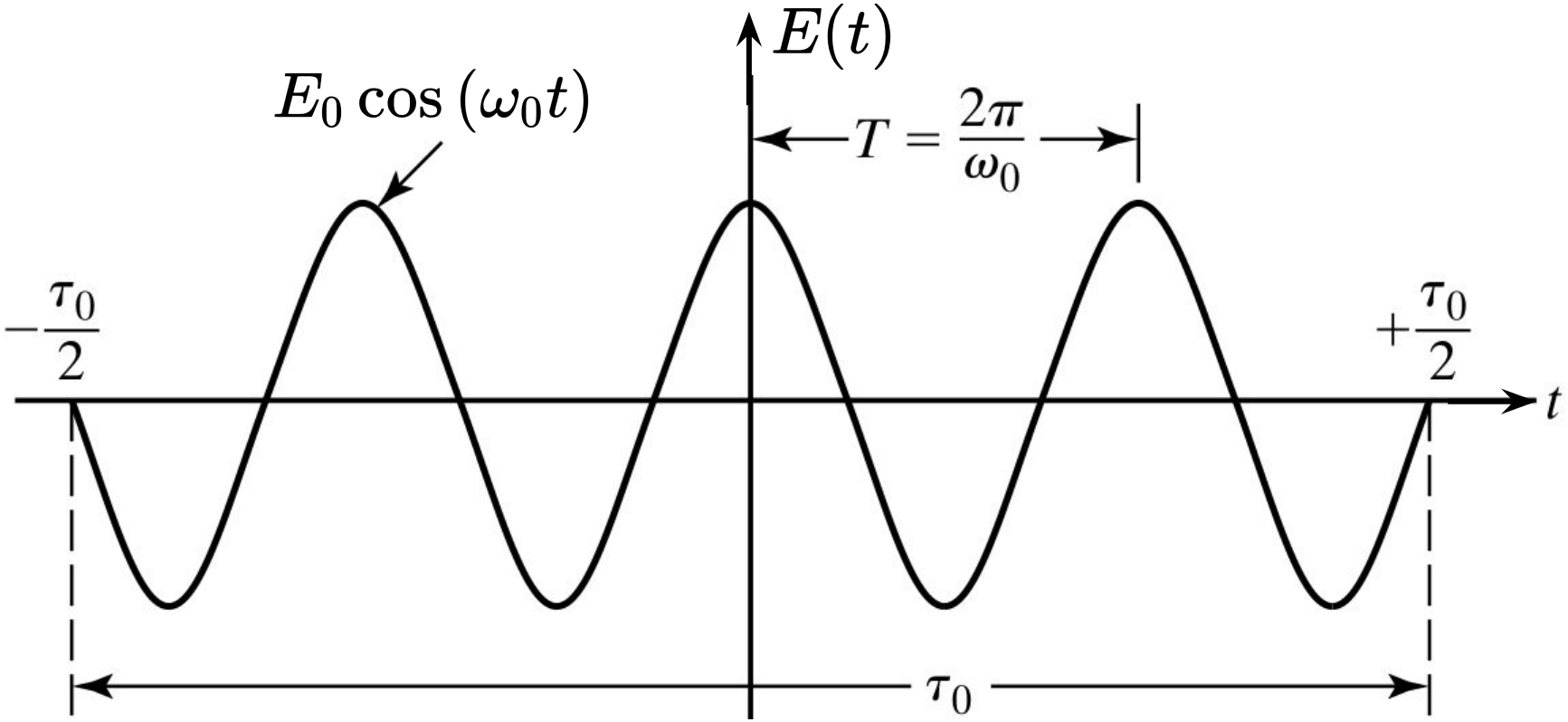
$$\boxed{\begin{aligned}\tilde{E}(\omega) &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} E(t)\exp(-i\omega t)dt \\ E(t) &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} \tilde{E}(\omega)\exp(i\omega t)d\omega\end{aligned}}$$

$\longleftarrow$ This is as per the definition of Fourier transform

Fourier transform of a harmonic wavetrain of finite duration

- Consider a finite wavetrain propagating in z-direction. The electric field of this wavetrain at a fixed position, say at $z = 0$, may be written as

$$\boxed{E(t) = \begin{cases} E_0 \exp(i\omega_0 t), & t \leqslant \left|\frac{\tau_0}{2}\right| \\ 0, & \text{elsewhere} \end{cases}}$$



Fourier transform of a harmonic wavetrain of finite duration

$$\tilde{E}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} E(t) \exp(-i\omega t) dt$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\frac{\tau_0}{2}}^{\frac{\tau_0}{2}} E_0 \exp\left[i(\omega_0 - \omega)t\right] d\omega + 0$$

$$= \sqrt{\frac{2}{\pi}} E_0 \left[\frac{\sin\left\{\frac{1}{2}(\omega_0 - \omega)\tau_0\right\}}{(\omega_0 - \omega)} \right]$$

$$= \sqrt{\frac{2}{\pi}} \frac{E_0}{\omega_0} \left[\frac{\sin\left\{\gamma(\Omega - 1)\right\}}{(\Omega - 1)} \right]$$

$$\tilde{E}(\omega) = \sqrt{\frac{2}{\pi}} \frac{E_0}{\omega_0} f(\Omega)$$

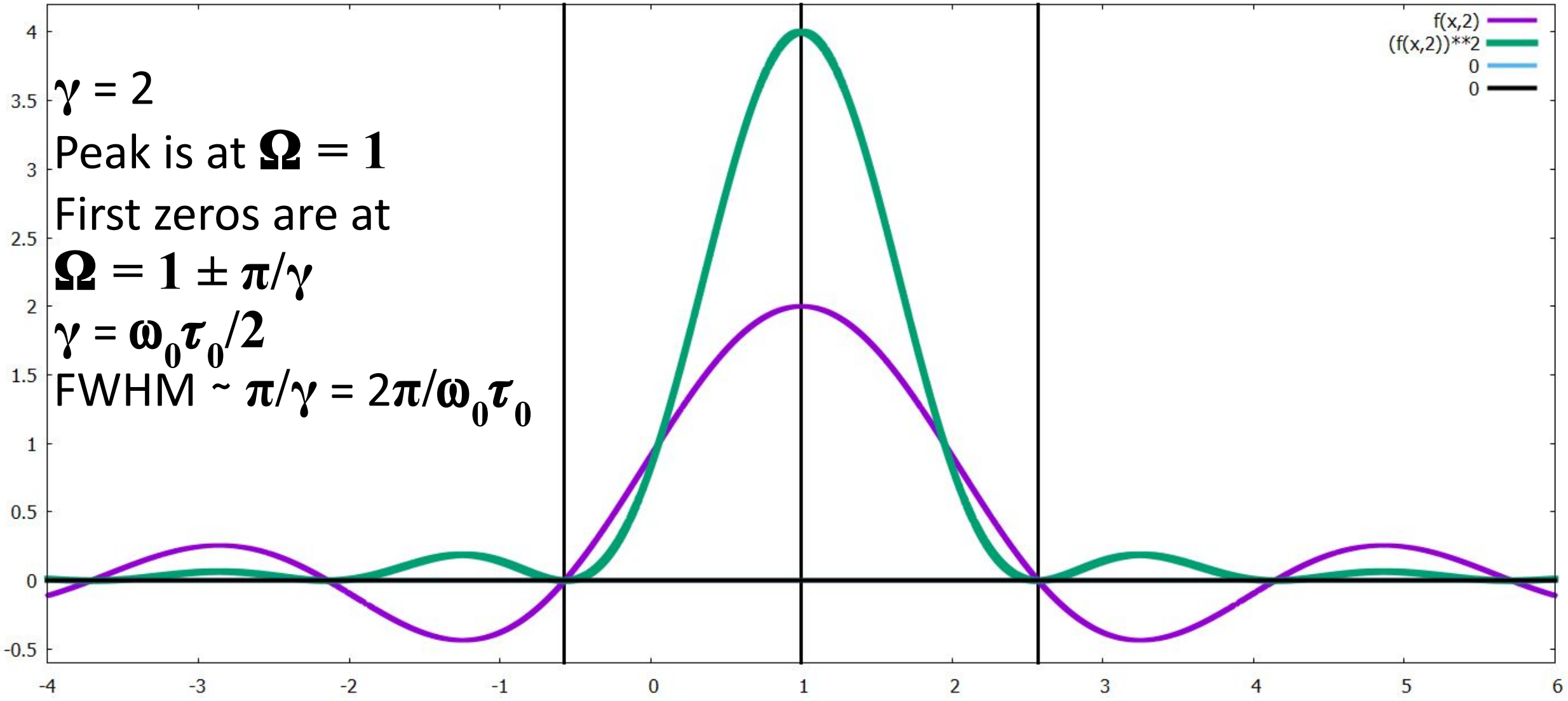
We define

$$\Omega = \frac{\omega}{\omega_0}$$

$$\gamma = \frac{\omega_0 \tau_0}{2}$$

$$f(\Omega) = \frac{\sin\left[\gamma(\Omega - 1)\right]}{(\Omega - 1)}$$

Plot of $f(\Omega)$ and $|f(\Omega)|^2$



$\gamma = 2$

Peak is at $\boldsymbol{\Omega = 1}$

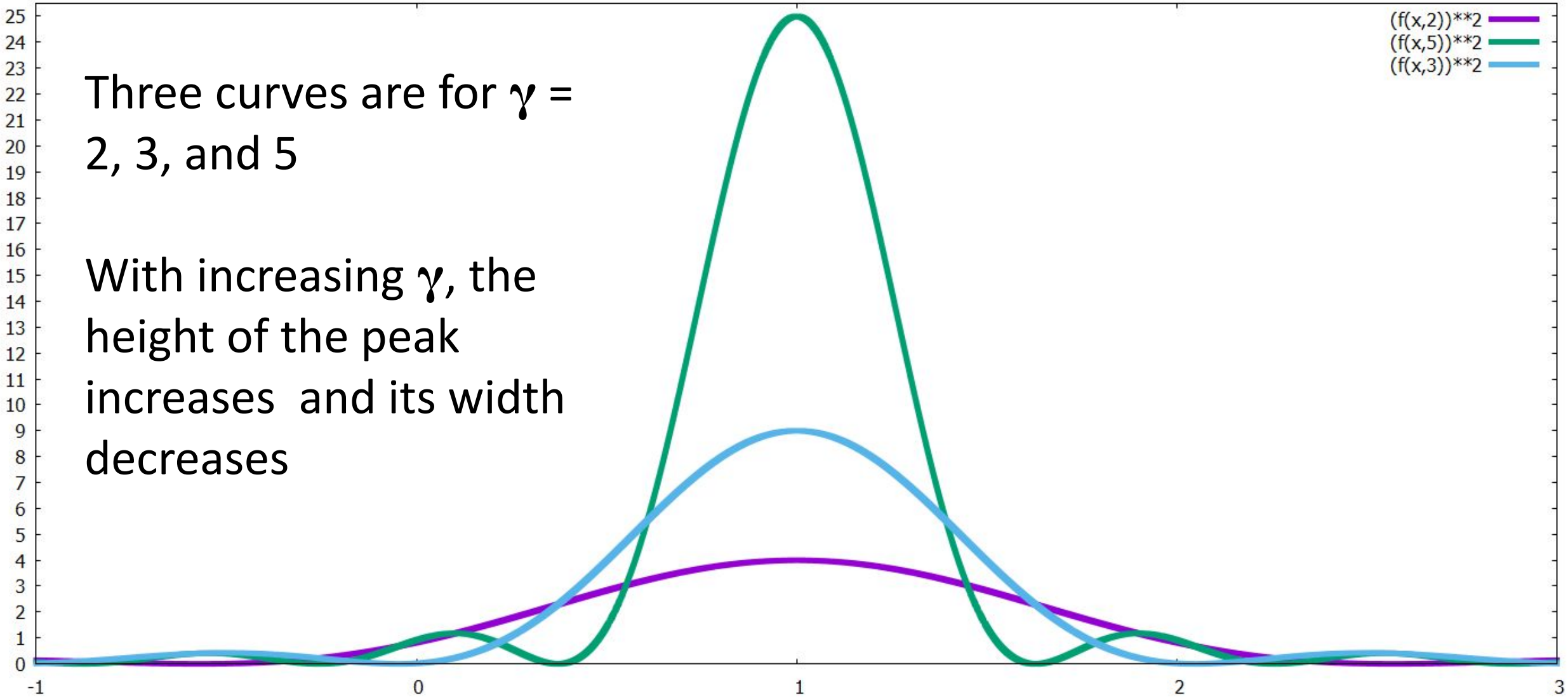
First zeros are at

$\boldsymbol{\Omega = 1 \pm \pi/\gamma}$

$\boldsymbol{\gamma = \omega_0 \tau_0 / 2}$

FWHM ~ $\boldsymbol{\pi/\gamma = 2\pi/\omega_0 \tau_0}$

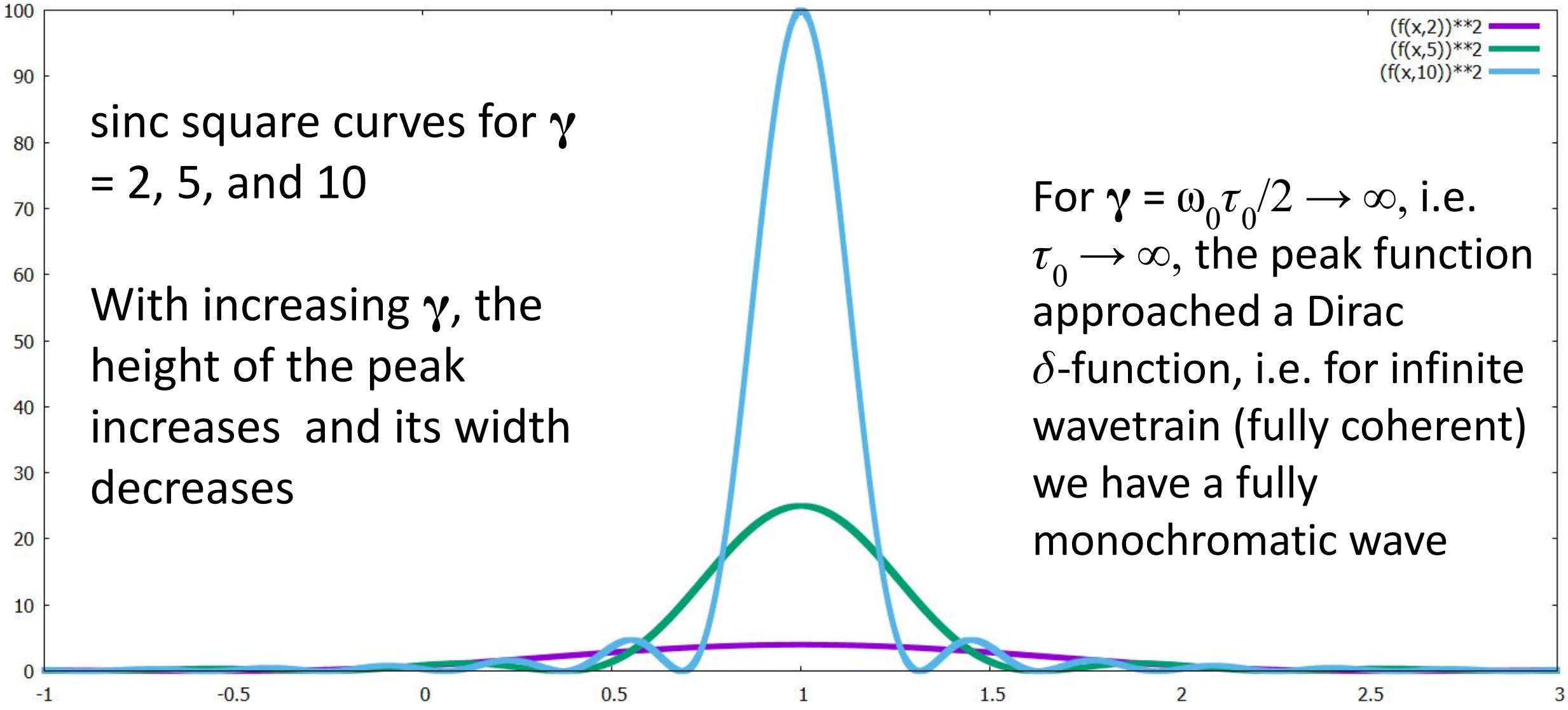
Plot of $|f(\Omega)|^2$



Three curves are for γ = 2, 3, and 5

With increasing γ, the height of the peak increases and its width decreases

Plot of $|f(\Omega)|^2$



sinc square curves for γ = 2, 5, and 10

With increasing γ, the height of the peak increases and its width decreases

For $\gamma = \omega_0\tau_0/2 \rightarrow \infty$, i.e. $\tau_0 \rightarrow \infty$, the peak function approached a Dirac δ-function, i.e. for infinite wavetrain (fully coherent) we have a fully monochromatic wave

Spectral width is inversely proportional to the time duration of the wavetrain

- We have obtained FWHM ($\boldsymbol{\Delta\Omega}$) ~ $\pi/\boldsymbol{\gamma}$; Also $\boldsymbol{\Omega} = \boldsymbol{\omega}/\boldsymbol{\omega}_0$ and $\boldsymbol{\gamma} = \omega_0\tau_0/2$
- So, $\boldsymbol{\Delta\omega}$ ~ $2\pi/\tau_0$
- Linear frequency $\nu = \boldsymbol{\omega}/2\pi \Rightarrow \boldsymbol{\Delta\nu}$ ~ $1/\tau_0$ or τ_0 ~ $1/\boldsymbol{\Delta\nu}$
- Coherence time ~ $\boldsymbol{\tau_0}$ is inversely proportional to the spectral width
- We have $\nu = c/\lambda \Rightarrow |\boldsymbol{\Delta\nu}| = c\,|\boldsymbol{\Delta\lambda}|/\lambda^2$
- Coherence time $\tau_c = \lambda^2/(c\boldsymbol{\Delta\lambda})$
- Coherence length $l_c = c\tau_c = \lambda^2/\boldsymbol{\Delta\lambda}$

Example: coherence length and coherence time for a few typical light sources

<u>Visible light (400-700 nm)</u>:

Spectral width $\Delta\lambda$ = 300 nm and mean wavelength λ_m = 550 nm

Coherence length $l_c = \lambda_m^2/\Delta\lambda \approx 1\ \mu m$

Coherence time $\tau_c = l_c/c = \lambda_m^2/(c\Delta\lambda) \approx 3$ fs (1 fs = 10^{-15} s)

<u>LED light (cosider a typical red LED)</u>:

Spectral width $\Delta\lambda \approx 20$ nm and mean wavelength λ_m = 600 nm

Coherence length $l_c = \lambda_m^2/\Delta\lambda \approx 18\ \mu m$

Coherence time $\tau_c = l_c/c = \lambda_m^2/(c\Delta\lambda) \approx 60$ fs

Example: coherence length and coherence time for a few typical light sources

<u>Gas discharge tube (take sodium vapor lamp)</u>:

For sodium D1 line, spectral width $\Delta\lambda \approx 0.01$ nm and wavelength $\lambda = 589$ nm

Coherence length $l_c = \lambda^2/\Delta\lambda \approx 3.5$ cm

Coherence time $\tau_c = l_c/c = \lambda^2/(c\Delta\lambda) \approx 0.1$ ns (1 ns = 10^{-9} s)

<u>Laser light (cosider a typical He-Ne laser)</u>:

Spectral width $\Delta\lambda \approx 10^{-6}$ nm and wavelength $\lambda = 632.8$ nm

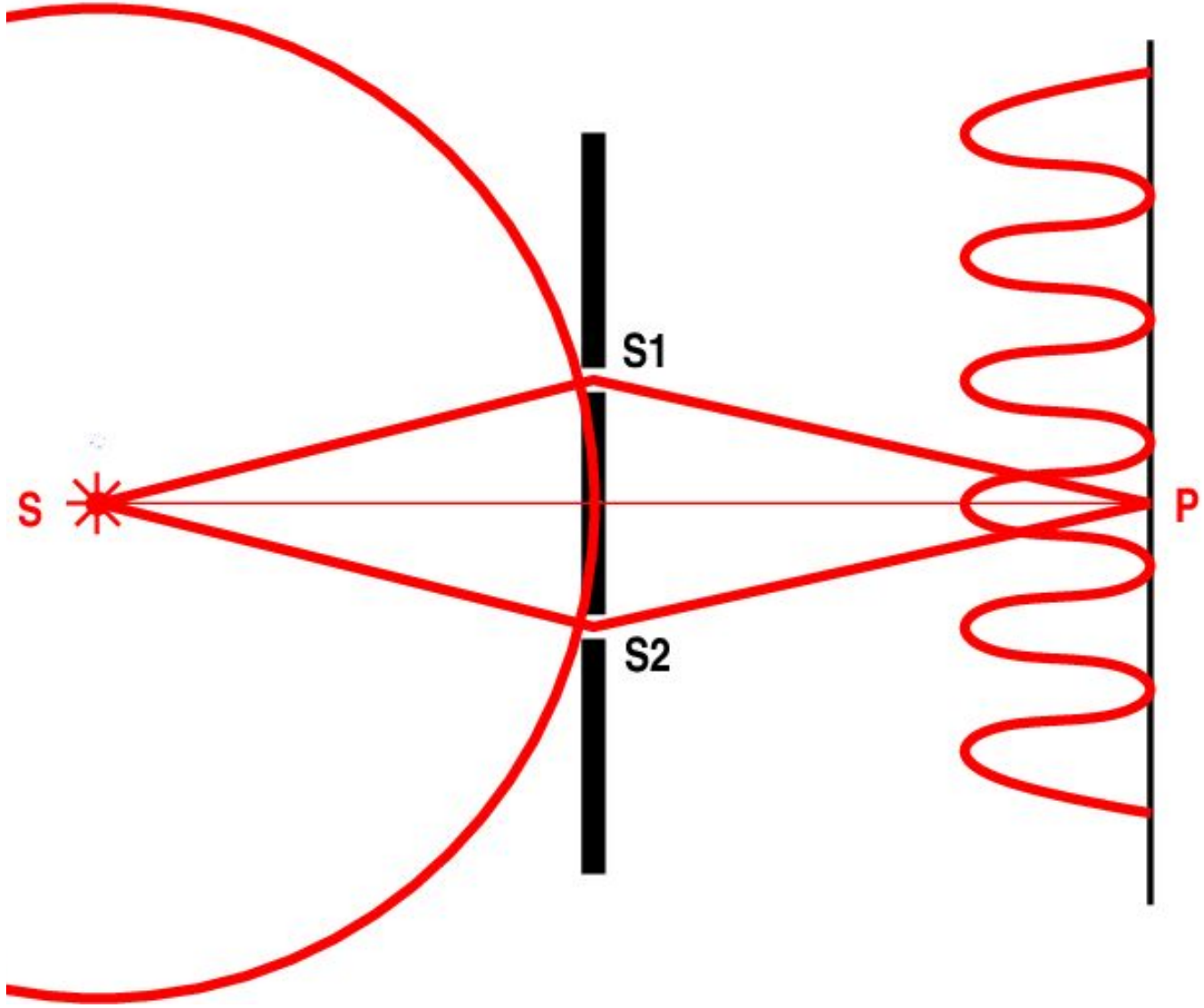
Coherence length $l_c = \lambda_m^2/\Delta\lambda \approx 400$ m

Coherence time $\tau_c = l_c/c = \lambda_m^2/(c\Delta\lambda) \approx 1.3$ μs

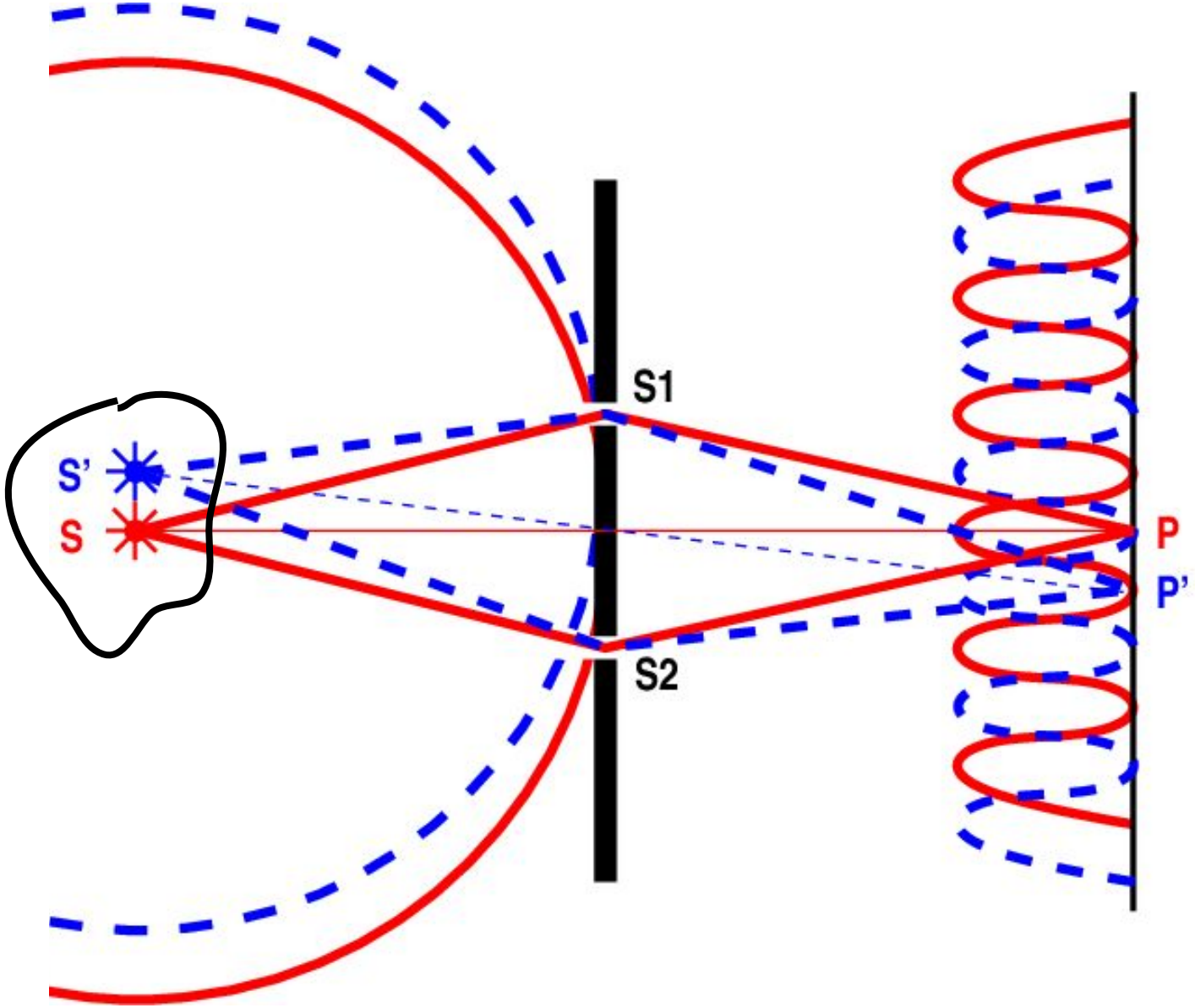
Measuring temporal coherence

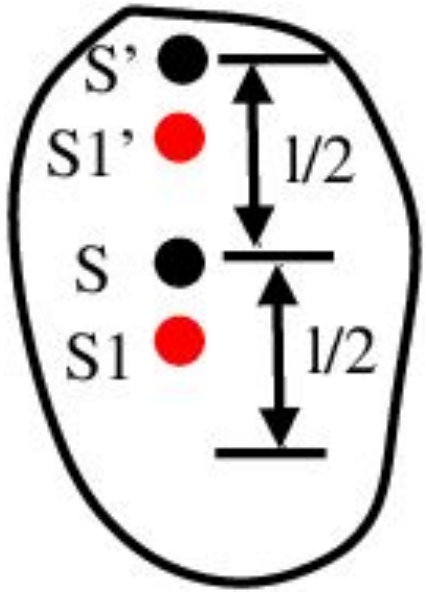
- Effect of temporal coherence become visible in interference experiments with division of amplitude (such as Michelson interferometer)
- Michelson interferometer setup can be used to measure temporal coherence of light from a given source

Spatial coherence

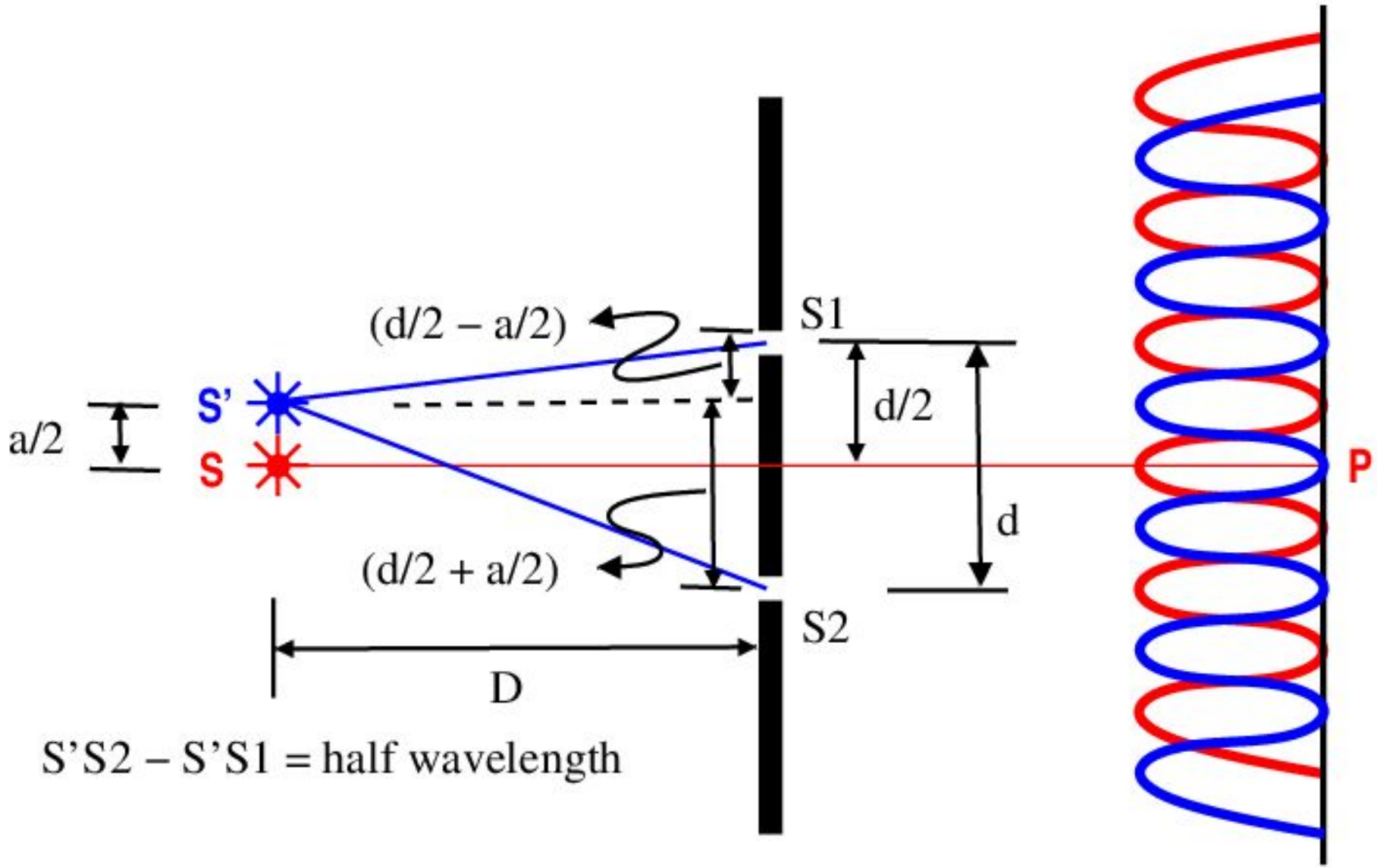


Spatial coherence





Expression for spatial coherence length



S'S2 – S'S1 = half wavelength

Expression for spatial coherence length

- $S'S1 = \sqrt{D^2 + (d/2 - a/2)^2}$
- We assume D >> d and D>> a
- Then,

$$S'S1 = D\left[1 + \left(\frac{d-a}{2D}\right)^2\right]^{1/2}$$

$$\approx D\left[1 + \frac{1}{2}\left(\frac{d-a}{2D}\right)^2\right]$$

$$\Rightarrow S'S1 \approx D + \frac{(d-a)^2}{8D}$$

Similarly,

$$S'S2 = D\left[1 + \left(\frac{d+a}{2D}\right)^2\right]^{1/2}$$

$$\approx D\left[1 + \frac{1}{2}\left(\frac{d+a}{2D}\right)^2\right]$$

$$\Rightarrow S'S2 \approx D + \frac{(d+a)^2}{8D}$$

- So, $S'S2 - S'S1 \approx \frac{1}{8D}\left[(d+a)^2 - (d-a)^2\right] = \frac{da}{2D}$

Expression for spatial coherence length

- For a to be spatial coherence length (interference pattern to be washed out), we should have $S'S2 - S'S1 = \frac{da}{2D} = \frac{\lambda}{2}$ i. e. $\boxed{a = \frac{\lambda D}{d}}$
- **Note that spatial coherence is tested at the plane of the slit and not at the plane of the screen**
- A circular disk of area $\pi a^2/4$ defines coherence area, and if l is the temporal coherence length, then a cylindrical volume $\pi a^2 l/4$ defines coherence volume
- We see that a increases when D increases and a decreases when d increases
- So, if a source of a given size is moved farther away its spatial coherence improves - this is because spherical wavefronts emitted by constituent point sources tend towards plane wave front at far away distance - so they reach the slits S1 and S2 nearly at the same phase - so spatial coherence improves

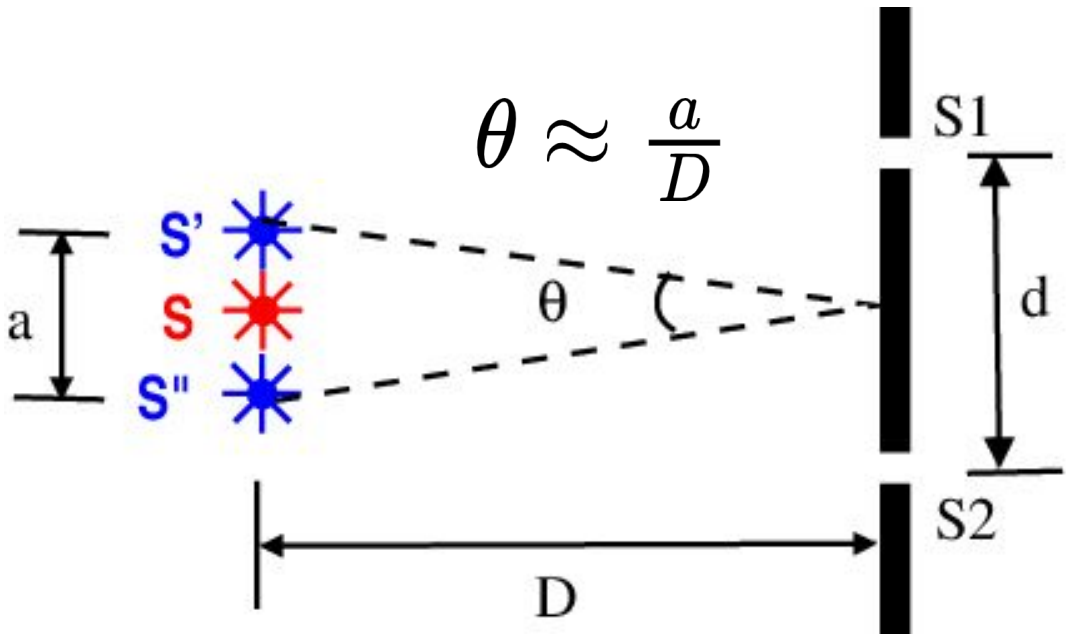
Spatial coherence improves at far away distance from broad source

https://www.youtube.com/watch?reload=9&v=4o48J4streE

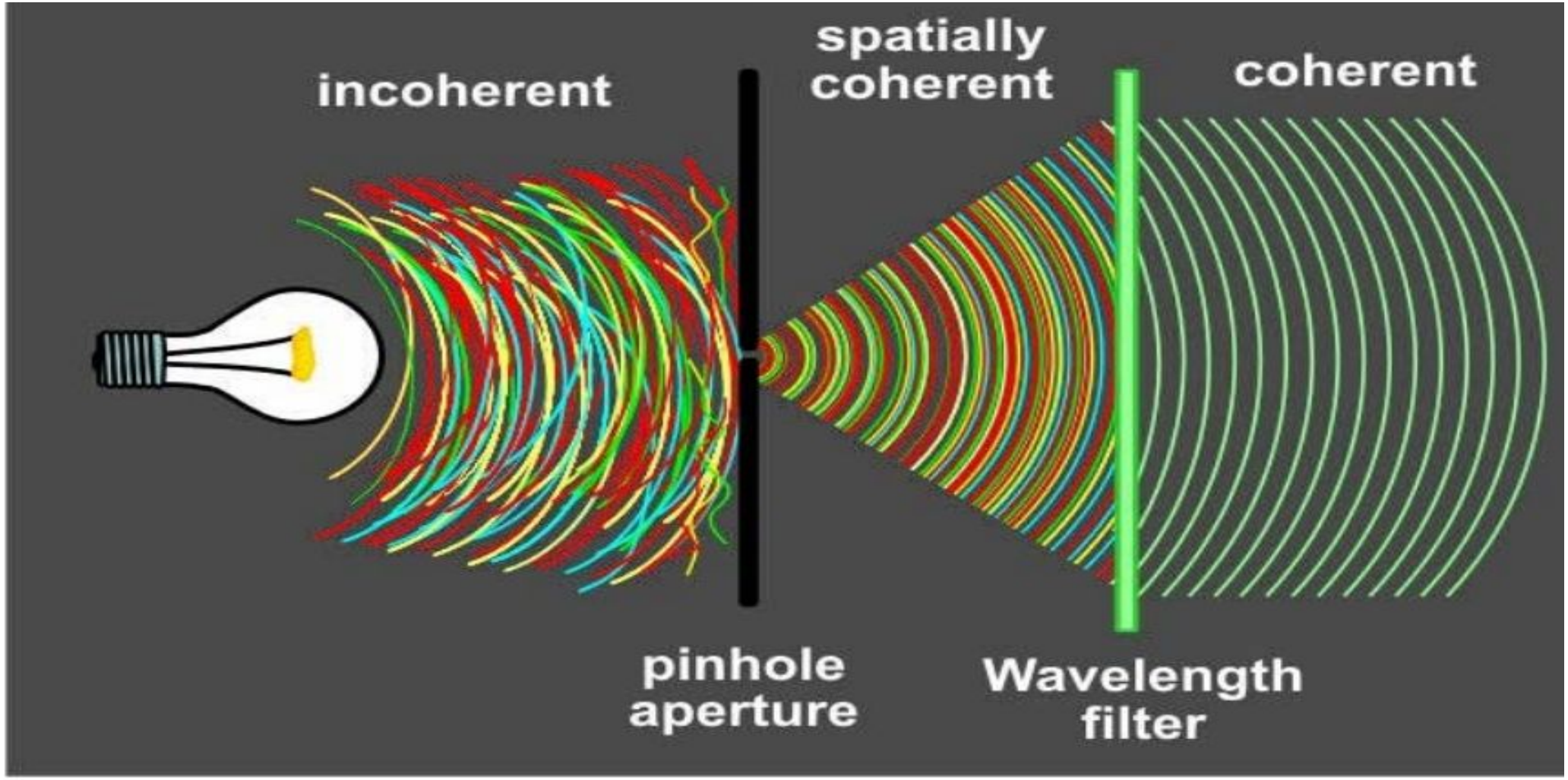
Spatial coherence size can be expressed in terms of an angle subtended by the source at the observation point

- The spatial coherence size of a source depends on the distance of the observation point from the source
- It can be conveniently expressed in terms of the angle subtended by the source at the observation point

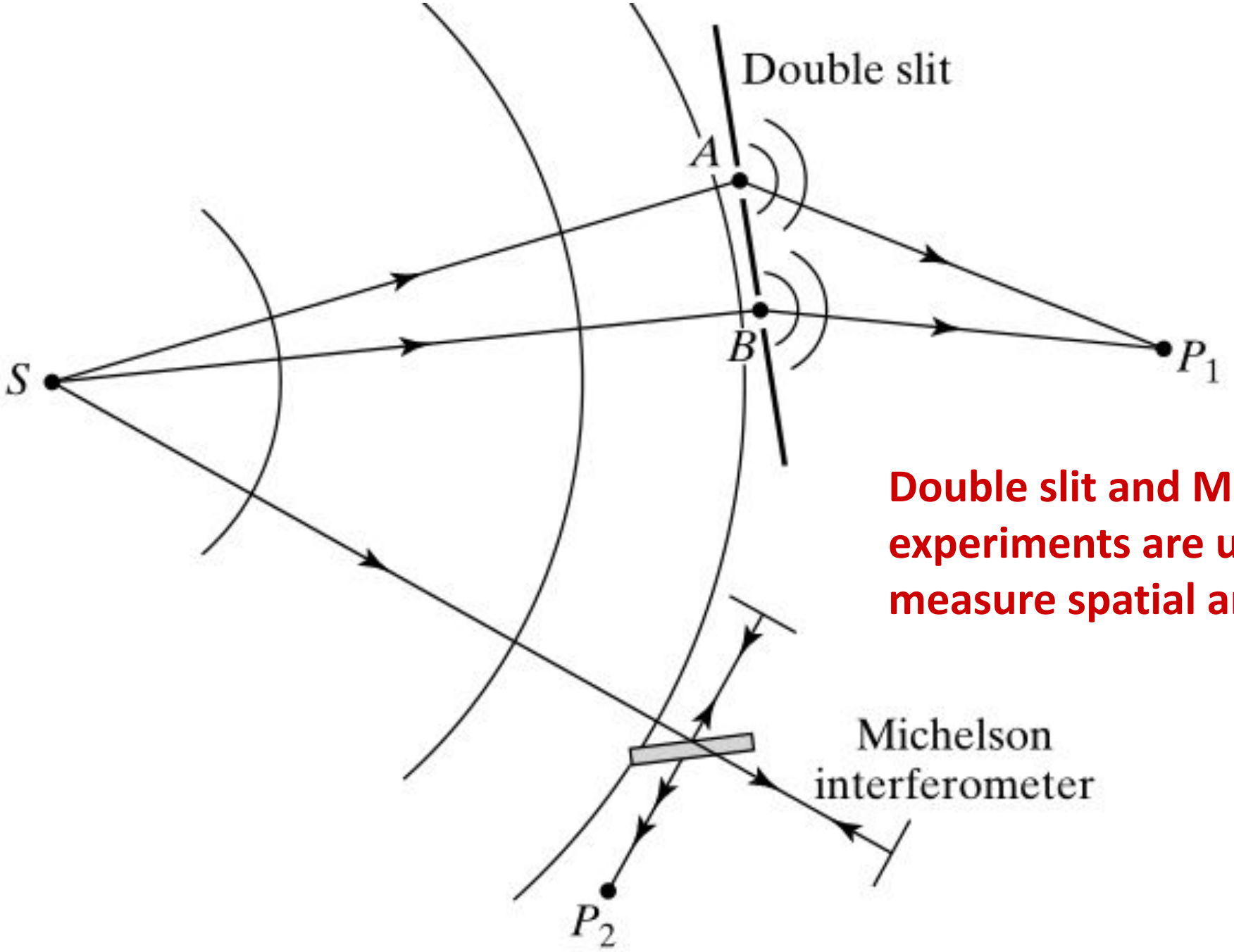
$$\theta \approx \frac{a}{D}$$



- We have got $a = \frac{\lambda D}{d} \Rightarrow d = \frac{\lambda D}{a} = \frac{\lambda}{\theta}$ **Note: θ is in radian**
- This means that an extended source subtending an angle θ at the slit plane, emitting light at an wavelength λ, can produce visible interference fringes in a double slit experiment, for a maximum slit separation of d.



Schematic picture showing how to get spatially and temporally coherent light from a light bulb, which is an extended source, giving spatially incoherent light, and it also gives polychromatic light, containing many wavelengths, which is temporally incoherent. We need to use a pinhole to improve spatial coherence and a wavelength filter to improve temporal coherence.



Double slit and Michelson interferometer experiments are used, respectively, to measure spatial and temporal coherence